CSE 482 Project – Part I: Data Collection and Preprocessing

A. Title and Team Members

Title: Movie Recommendation System using Collaborative Filtering
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Note: This project is completed individually.

B. Data Collection and Storage

This project focuses on the movie recommendation domain, addressing the problem of information overload, where users are presented with a large number of choices.

The dataset used is the MovieLens latest-small dataset from GroupLens Research. It contains:
- 100,836 ratings
- 610 users
- 9,742 movies
- Ratings on a 0.5–5.0 scale

The dataset was originally provided as CSV files:
- ratings.csv: userId, movieId, rating, timestamp
- movies.csv: movieId, title, genres

The data was first loaded and processed in Python using the Pandas library.

After preprocessing, the cleaned data was stored in a MySQL relational database running in a Docker container. The database (cse482_movielens) contains the following tables:
- ratings: user–movie rating interactions
- movies: movie metadata (titles and genres)
- movies_genres: one-hot encoded genre features
- ratings_with_movies: merged dataset for analysis

This approach combines structured data storage with efficient querying using SQL.

C. Data Preprocessing

Although the dataset is relatively clean, several preprocessing steps were required to ensure it is suitable for recommendation algorithms.

1. Data Quality Assessment:
- No missing values in ratings, movies, or tags
- 8 missing values in tmdbId in links (not used)
- No duplicate records found
- Rating values are valid within the range 0.5 to 5.0

2. Data Consistency Checks:
- Verified that all rated movies exist in movies.csv
- Identified 18 movies without ratings
- Confirmed consistency of user and movie identifiers across datasets

3. Feature Engineering:
- Converted timestamps to human-readable datetime format
- Merged ratings and movies to create a unified dataset
- Extracted genre information and created 20 one-hot encoded genre features

4. Statistical Analysis:
- Average rating: 3.50
- Median rating: 3.5
- Ratings per user: minimum 20, maximum 2698
- Ratings per movie: minimum 1, maximum 329

5. Data Sparsity Analysis:
- Total possible user–movie pairs: 5,931,640
- Observed ratings: 100,836
- Density: 0.017
- Sparsity: 0.983

This high sparsity justifies the use of collaborative filtering techniques.

D. Report

This project aims to develop a movie recommendation system using collaborative filtering techniques, specifically user-based and item-based nearest neighbor methods.

The dataset is a structured, record-based dataset where each record represents a user's rating of a movie. The data spans from 1996 to 2018 and captures explicit user preferences.

Data preprocessing involved cleaning, validation, transformation, and feature extraction. The dataset was then stored in a MySQL database using Docker, enabling efficient storage and retrieval using SQL queries.

Example SQL queries were used to:
- Identify the most frequently rated movies
- Analyze rating distributions
- Determine user activity levels

These steps ensure that the data is properly structured and ready for the implementation of recommendation algorithms.

The processed dataset and database structure provide a strong foundation for Part II, where collaborative filtering methods will be applied and evaluated.